

```

# Install Required Libraries
# Install Scanpy, Squidpy, and supporting visualization libraries
!pip install scanpy squidpy anndata matplotlib seaborn pooch

# Install Leiden algorithm package and graph dependencies
!pip install leidenalg igraph

# Upgrade Scanpy-related graph libraries to avoid compatibility issues
!pip install scanpy squidpy leidenalg igraph --upgrade

Requirement already satisfied: wrapt>=1.12.0 in /usr/local/lib/python3.12/dist-packages (from omnipath>=1.0.7->squidpy) (1.17.0)
Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.12/dist-packages (from pandas>=2.3->scanpy) (2025.2)
Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.12/dist-packages (from pandas>=2.3->scanpy) (2026.1)
Requirement already satisfied: platformdirs>=2.5.0 in /usr/local/lib/python3.12/dist-packages (from pooch>=1.6->squidpy) (4.9.0)
Requirement already satisfied: imageio!=2.35.0,>=2.33 in /usr/local/lib/python3.12/dist-packages (from scikit-image>=0.25->squidpy) (2.36.0)
Requirement already satisfied: lazy-loader>=0.4 in /usr/local/lib/python3.12/dist-packages (from scikit-image>=0.25->squidpy) (0.4.0)
Requirement already satisfied: threadpoolctl>=3.1.0 in /usr/local/lib/python3.12/dist-packages (from scikit-learn>=1.6->scanpy) (3.6.0)
Requirement already satisfied: annsel>=0.1.2 in /usr/local/lib/python3.12/dist-packages (from spatialdata>=0.7.1->squidpy) (0.1.2)
Requirement already satisfied: datashader in /usr/local/lib/python3.12/dist-packages (from spatialdata>=0.7.1->squidpy) (0.19.0)
Requirement already satisfied: distributed<2026.1.2 in /usr/local/lib/python3.12/dist-packages (from spatialdata>=0.7.1->squidpy) (2025.12.0)
Requirement already satisfied: geopandas>=0.14 in /usr/local/lib/python3.12/dist-packages (from spatialdata>=0.7.1->squidpy) (0.14.3)
Requirement already satisfied: multiscale-spatial-image==2.0.3 in /usr/local/lib/python3.12/dist-packages (from spatialdata>=0.7.1->squidpy) (2.0.3)
Requirement already satisfied: ome-zarr>=0.14.0 in /usr/local/lib/python3.12/dist-packages (from spatialdata>=0.7.1->squidpy) (0.14.0)
Requirement already satisfied: pyarrow in /usr/local/lib/python3.12/dist-packages (from spatialdata>=0.7.1->squidpy) (18.1.0)
Requirement already satisfied: rich in /usr/local/lib/python3.12/dist-packages (from spatialdata>=0.7.1->squidpy) (13.9.4)
Requirement already satisfied: setuptools in /usr/local/lib/python3.12/dist-packages (from spatialdata>=0.7.1->squidpy) (75.2.0)
Requirement already satisfied: shapely>=2.0.1 in /usr/local/lib/python3.12/dist-packages (from spatialdata>=0.7.1->squidpy) (2.0.6)
Requirement already satisfied: spatial-image>=1.2.3 in /usr/local/lib/python3.12/dist-packages (from spatialdata>=0.7.1->squidpy) (1.2.3)
Requirement already satisfied: universal-pathlib>=0.2.6 in /usr/local/lib/python3.12/dist-packages (from spatialdata>=0.7.1->squidpy) (0.2.6)
Requirement already satisfied: xarray-spatial>=0.3.5 in /usr/local/lib/python3.12/dist-packages (from spatialdata>=0.7.1->squidpy) (0.3.5)
Requirement already satisfied: xarray-dataclass>=3.0.0 in /usr/local/lib/python3.12/dist-packages (from multiscale-spatial-image>=2.0.3->squidpy) (3.0.0)
Requirement already satisfied: donfig>=0.8 in /usr/local/lib/python3.12/dist-packages (from zarr>=3->squidpy) (0.8.1.post1)
Requirement already satisfied: google-crc32c>=1.5 in /usr/local/lib/python3.12/dist-packages (from zarr>=3->squidpy) (1.8.0)
Requirement already satisfied: numcodecs>=0.14 in /usr/local/lib/python3.12/dist-packages (from zarr>=3->squidpy) (0.16.5)
Requirement already satisfied: more-itertools>=10.7 in /usr/local/lib/python3.12/dist-packages (from annsel>=0.1.2->spatialdata) (10.7.0)
Requirement already satisfied: narwhals>=2.14 in /usr/local/lib/python3.12/dist-packages (from narwhals[pandas]>=2.14->annsel) (2.14.0)
Requirement already satisfied: jinja2>=2.10.3 in /usr/local/lib/python3.12/dist-packages (from distributed<2026.1.2->spatialdata) (3.1.4)
Requirement already satisfied: locket>=1.0.0 in /usr/local/lib/python3.12/dist-packages (from distributed<2026.1.2->spatialdata) (1.0.0)
Requirement already satisfied: msgpack>=1.0.2 in /usr/local/lib/python3.12/dist-packages (from distributed<2026.1.2->spatialdata) (1.0.8)
Requirement already satisfied: psutil>=5.8.0 in /usr/local/lib/python3.12/dist-packages (from distributed<2026.1.2->spatialdata) (5.8.0)
Requirement already satisfied: sortedcontainers>=2.0.5 in /usr/local/lib/python3.12/dist-packages (from distributed<2026.1.2->spatialdata) (2.0.5)
Requirement already satisfied: tblib!=3.2.0,!>3.2.1,>=1.6.0 in /usr/local/lib/python3.12/dist-packages (from distributed<2026.1.2->spatialdata) (3.2.0)
Requirement already satisfied: tornado>=6.2.0 in /usr/local/lib/python3.12/dist-packages (from distributed<2026.1.2->spatialdata) (6.4.0)
Requirement already satisfied: zict>=3.0.0 in /usr/local/lib/python3.12/dist-packages (from distributed<2026.1.2->spatialdata) (3.0.0)
Requirement already satisfied: pyogrio>=0.7.2 in /usr/local/lib/python3.12/dist-packages (from geopandas>=0.14->spatialdata) (0.7.2)
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Requirement already satisfied: typeguard>=4.0.1 in /usr/local/lib/python3.12/dist-packages (from inflect>=4.1.0->omnipath>=1.0.7) (4.0.1)
Requirement already satisfied: rangehttpserver in /usr/local/lib/python3.12/dist-packages (from ome-zarr>=0.14.0->spatialdata) (1.0.0)
Requirement already satisfied: Deprecated in /usr/local/lib/python3.12/dist-packages (from ome-zarr>=0.14.0->spatialdata) (1.2.14)
Requirement already satisfied: slicerator>=0.9.8 in /usr/local/lib/python3.12/dist-packages (from pims>=0.4.1->dask-image>=0.5.0) (0.9.8)
Requirement already satisfied: charset-normalizer<4,>=2 in /usr/local/lib/python3.12/dist-packages (from requests>=2.24.0->omnipath>=1.0.7) (3.4.0)
Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.12/dist-packages (from requests>=2.24.0->omnipath>=1.0.7) (3.10.0)
Requirement already satisfied: pathlib-abc<0.6.0,>=0.5.1 in /usr/local/lib/python3.12/dist-packages (from universal-pathlib>=0.2.6->spatialdata) (0.5.1)
Requirement already satisfied: zstandard in /usr/local/lib/python3.12/dist-packages (from xarray-spatial>=0.3.5->spatialdata) (0.23.0)
Requirement already satisfied: colorcet in /usr/local/lib/python3.12/dist-packages (from datashader->spatialdata>=0.7.1->squidpy) (2.10.0)
Requirement already satisfied: multipledispatch in /usr/local/lib/python3.12/dist-packages (from datashader->spatialdata>=0.7.1->squidpy) (0.6.0)
Requirement already satisfied: param in /usr/local/lib/python3.12/dist-packages (from datashader->spatialdata>=0.7.1->squidpy) (2.0.2)
Requirement already satisfied: pyct in /usr/local/lib/python3.12/dist-packages (from datashader->spatialdata>=0.7.1->squidpy) (0.6.0)
Requirement already satisfied: s3fs in /usr/local/lib/python3.12/dist-packages (from fsspec[http,s3]->spatialdata>=0.7.1->squidpy) (2025.10.0)
Requirement already satisfied: markdown-it-py>=2.2.0 in /usr/local/lib/python3.12/dist-packages (from rich->spatialdata>=0.7.1) (3.0.0)
Requirement already satisfied: pygments<3.0.0,>=2.13.0 in /usr/local/lib/python3.12/dist-packages (from rich->spatialdata>=0.7.1) (2.18.0)
Requirement already satisfied: MarkupSafe>=2.0 in /usr/local/lib/python3.12/dist-packages (from jinja2>=2.10.3->distributed<2026.1.2->spatialdata) (3.0.2)
Requirement already satisfied: mdurl~=0.1 in /usr/local/lib/python3.12/dist-packages (from markdown-it-py>=2.2.0->rich->spatialdata) (0.1.2)
Requirement already satisfied: aiobotocore<3.0.0,>=2.5.4 in /usr/local/lib/python3.12/dist-packages (from s3fs->fsspec[http,s3]) (2.14.0)
Requirement already satisfied: aioitertools<1.0.0,>=0.5.1 in /usr/local/lib/python3.12/dist-packages (from aiobotocore<3.0.0,>=2.5.4) (0.11.0)
Requirement already satisfied: botocore<1.41.6,>=1.41.0 in /usr/local/lib/python3.12/dist-packages (from aiobotocore<3.0.0,>=2.5.4) (1.41.0)
Requirement already satisfied: jmespath<2.0.0,>=0.7.1 in /usr/local/lib/python3.12/dist-packages (from aiobotocore<3.0.0,>=2.5.4) (1.0.1)

```

```

# Import libraries
import scanpy as sc
import squidpy as sq
import matplotlib.pyplot as plt
import numpy as np

```

```

# Load Example Single-Cell Dataset
adata = sc.datasets.pbmc3k_processed()

```

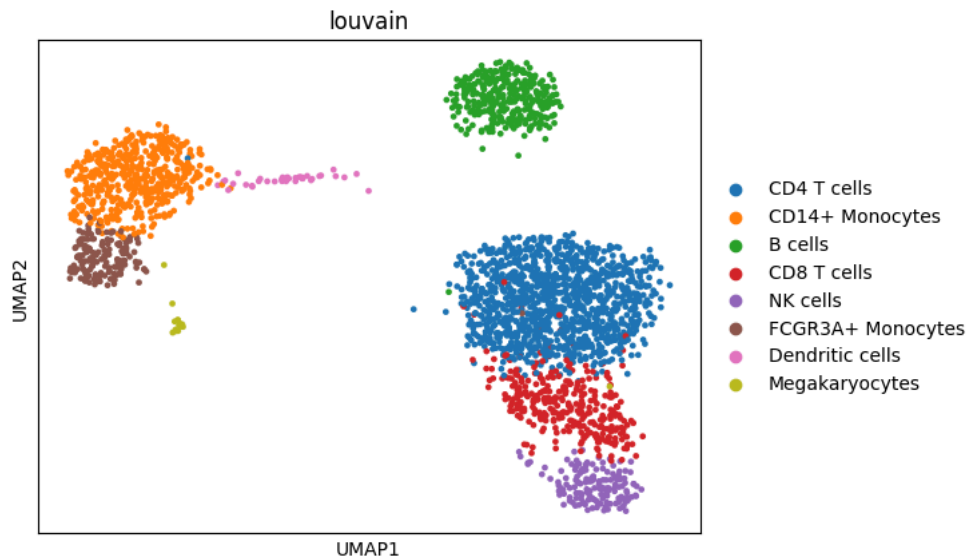
100%

23.5M/23.5M [00:00<00:00, 62.5MB/s]

```
# View data
adata
```

```
AnnData object with n_obs × n_vars = 2638 × 1838
  obs: 'n_genes', 'percent_mito', 'n_counts', 'louvain'
  var: 'n_cells'
  uns: 'draw_graph', 'louvain', 'louvain_colors', 'neighbors', 'pca', 'rank_genes_groups'
  obsm: 'X_pca', 'X_tsne', 'X_umap', 'X_draw_graph_fr'
  varm: 'PCs'
  obsp: 'distances', 'connectivities'
```

```
# Basic Visualization of Cell Clusters
# Plot UMAP visualization colored by Louvain clusters
# UMAP reduces high-dimensional data into 2D space
sc.pl.umap(adata, color="louvain")
```



```
# Explore Dataset Structure
# Access expression matrix
adata.X
# View first few rows of observation metadata (cells information)
adata.obs.head()
# View first few rows of variable metadata (genes information)
adata.var.head()
```

	n_cells
index	
TNFRSF4	155
CPSF3L	202
ATAD3C	9
C1orf86	501
RER1	608

```
# Subset Example
# Select a subset of cells and genes
# First 100 cells and first 500 genes
adata_subset = adata[:100, :500]
```

```
# Display shape of subset dataset
adata_subset.shape
```

```
(100, 500)
```

```
# Load Spatial Transcriptomics Dataset (Visium Fluorescence Data)
```

```
# Load spatial transcriptomics dataset from Squidpy
adata = sq.datasets.visium_fluo_adata()
```

```
# Display dataset information
adata
```

```
INFO      Downloading visium_fluo_adata.h5ad from
```

```
https://exampledata.scverse.org/squidpy/figshare/visium\_fluo\_adata.h5ad
```

```
Downloading data from 'https://exampledata.scverse.org/squidpy/figshare/visium\_fluo\_adata.h5ad' to file '/content/data/anndata/v
100%
```

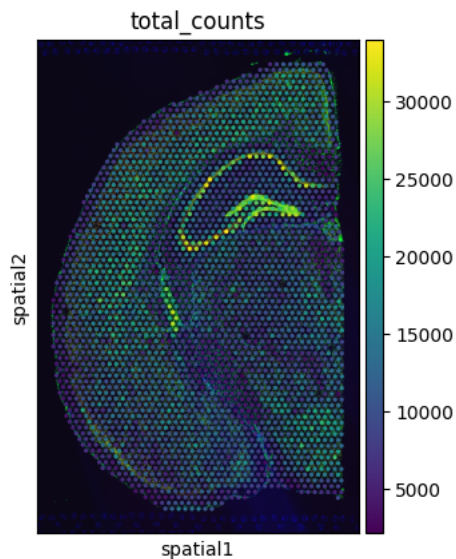
```
254M/254M
```

```
AnnData object with n_obs × n_vars = 2800 × 16562
```

```
  obs: 'in_tissue', 'array_row', 'array_col', 'n_genes_by_counts', 'log1p_n_genes_by_counts', 'total_counts',
'log1p_total_counts', 'pct_counts_in_top_50_genes', 'pct_counts_in_top_100_genes', 'pct_counts_in_top_200_genes',
'pct_counts_in_top_500_genes', 'total_counts_MT', 'log1p_total_counts_MT', 'pct_counts_MT', 'n_counts', 'leiden', 'cluster'
  var: 'gene_ids', 'feature_types', 'genome', 'MT', 'n_cells_by_counts', 'mean_counts', 'log1p_mean_counts',
'pct_dropout_by_counts', 'total_counts', 'log1p_total_counts', 'n_cells', 'highly_variable', 'highly_variable_rank', 'means',
'variances', 'variances_norm'
  uns: 'cluster_colors', 'hvg', 'leiden', 'leiden_colors', 'neighbors', 'pca', 'spatial', 'umap'
  obsm: 'X_pca', 'X_umap', 'spatial'
  varm: 'PCs'
  obsp: 'connectivities', 'distances'
```

```
# Spatial Visualization
# Plot spatial distribution of total counts across tissue spots
# Each dot represents a spatial location in the tissue
sc.pl.spatial(adata, color="total_counts")
```

```
/tmp/ipykernel_13391/3742325128.py:4: FutureWarning: Use `squidpy.pl.spatial_scatter` instead.
sc.pl.spatial(adata, color="total_counts")
```



```
# Preprocessing Spatial Transcriptomics Data
```

```
# Normalize total counts across spots/cells
# Makes expression values comparable
sc.pp.normalize_total(adata)
```

```
# Log-transform the normalized data
# Stabilizes variance and reduces skewness
sc.pp.log1p(adata)
```

```
# Perform Principal Component Analysis (PCA)
# Reduces dimensionality while keeping important variation
sc.pp.pca(adata)
```

```
# Compute neighborhood graph for clustering and UMAP
sc.pp.neighbors(adata)
```

```
# Compute UMAP embedding for visualization
```

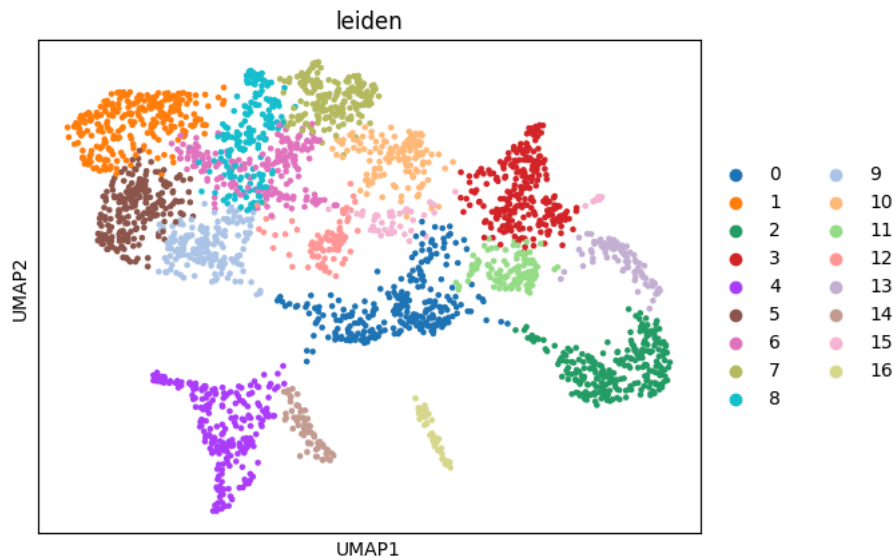
```
sc.tl.umap(adata)
```

```
# Clustering Analysis
```

```
# Perform Leiden clustering algorithm
# Groups similar cells/spots together
sc.tl.leiden(adata)
```

```
# Visualize UMAP colored by Leiden clusters
sc.pl.umap(adata, color="leiden")
```

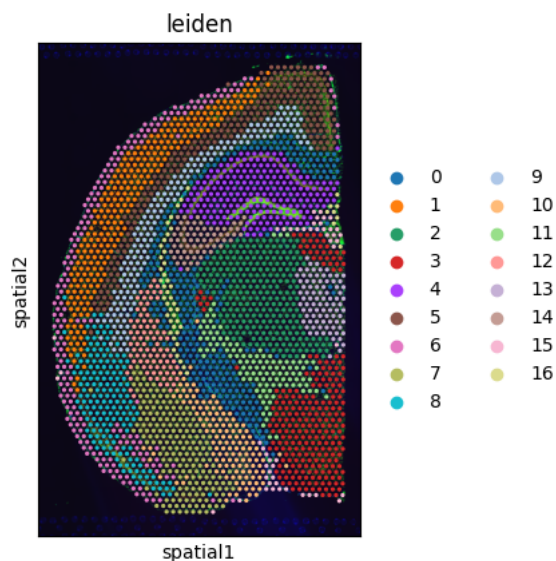
/tmp/ipykernel_13391/683407982.py:5: FutureWarning: The `igraph` implementation of leiden clustering is *orders of magnitude faster than the `leidenalg` implementation. In the future, the default backend for leiden will be igraph instead of leidenalg. To achieve the future defaults please pass: `use\_igraph=True`
 sc.tl.leiden(adata)



```
# Spatial Cluster Visualization
```

```
# Plot spatial locations colored by Leiden clusters
# Helps identify spatial organization of clusters
sc.pl.spatial(adata, color="leiden")
```

/tmp/ipykernel_13391/1950083337.py:4: FutureWarning: Use `squidpy.pl.spatial\_scatter` instead.
 sc.pl.spatial(adata, color="leiden")



```
# Load Another Spatial Dataset (Visium H&E data) (Human Neocortex Example)
# Load Visium Human Neocortex dataset
adata = sq.datasets.visium_hne_adata()
```

```
# Display dataset information
adata
```

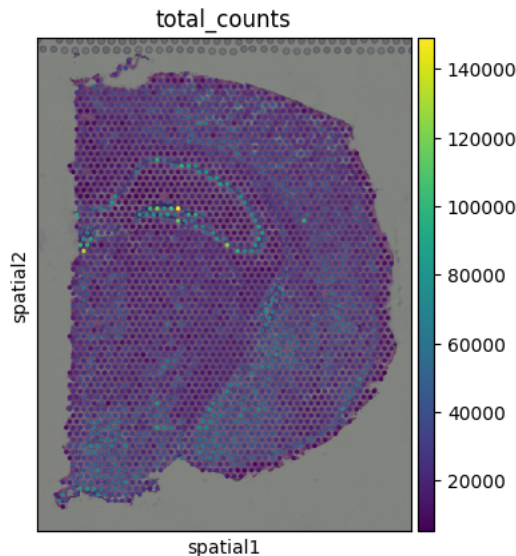
```
INFO Downloading visium_hne_adata.h5ad from
https://exampledata.scverse.org/squidpy/figshare/visium\_hne\_adata.h5ad
Downloading data from 'https://exampledata.scverse.org/squidpy/figshare/visium\_hne\_adata.h5ad' to file '/content/data/anndata/visium_hne_adata.h5ad'
100%

329M/329M

AnnData object with n_obs × n_vars = 2688 × 18078
  obs: 'in_tissue', 'array_row', 'array_col', 'n_genes_by_counts', 'log1p_n_genes_by_counts', 'total_counts',
'log1p_total_counts', 'pct_counts_in_top_50_genes', 'pct_counts_in_top_100_genes', 'pct_counts_in_top_200_genes',
'pct_counts_in_top_500_genes', 'total_counts_mt', 'log1p_total_counts_mt', 'pct_counts_mt', 'n_counts', 'leiden', 'cluster'
  var: 'gene_ids', 'feature_types', 'genome', 'mt', 'n_cells_by_counts', 'mean_counts', 'log1p_mean_counts',
'pct_dropout_by_counts', 'total_counts', 'log1p_total_counts', 'n_cells', 'highly_variable', 'highly_variable_rank', 'means',
'variances', 'variances_norm'
  uns: 'cluster_colors', 'hvg', 'leiden', 'leiden_colors', 'neighbors', 'pca', 'rank_genes_groups', 'spatial', 'umap'
  obsm: 'X_pca', 'X_umap', 'spatial'
  varm: 'PCs'
  obsp: 'connectivities', 'distances'
```

```
# Visualize Spatial Gene Counts
# Plot spatial distribution of total transcript counts
sc.pl.spatial(adata, color="total_counts")
```

```
/tmp/ipykernel_13391/3385223190.py:3: FutureWarning: Use `squidpy.pl.spatial_scatter` instead.
sc.pl.spatial(adata, color="total_counts")
```



```
# Preprocessing Pipeline Again for New Dataset
# Normalize total counts
sc.pp.normalize_total(adata)

# Apply log transformation
sc.pp.log1p(adata)

# Run PCA for dimensionality reduction
sc.pp.pca(adata)

# Compute nearest-neighbor graph
sc.pp.neighbors(adata)

# Generate UMAP embedding
sc.tl.umap(adata)

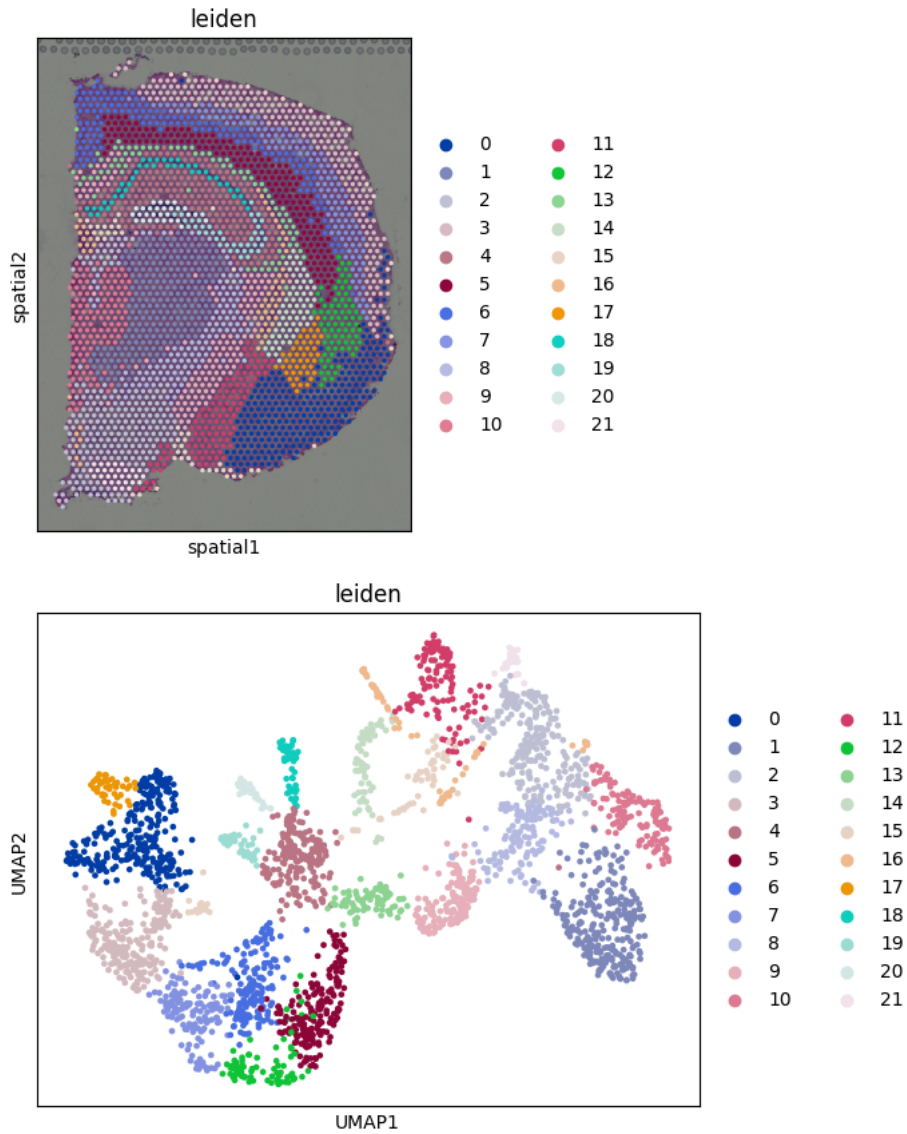
# Run Leiden clustering
sc.tl.leiden(adata)
```

```
/tmp/ipykernel_13391/3076487841.py:18: FutureWarning: The `igraph` implementation of leiden clustering is *orders of magnitude f
In the future, the default backend for leiden will be igraph instead of leidenalg. To achieve the future defaults please pass: `
sc.tl.leiden(adata)
```

```
# Visualize Clustering Results
# Plot spatial tissue map with Leiden clusters
sc.pl.spatial(adata, color="leiden")

# Plot UMAP visualization with Leiden clusters
sc.pl.umap(adata, color="leiden")
```

```
/tmp/ipykernel_13391/2259612200.py:3: FutureWarning: Use `squidpy.pl.spatial_scatter` instead.
sc.pl.spatial(adata, color="leiden")
```



```
# Xenium Dataset
```

```
import os
```

```
os.listdir("/content")
```

```
['.config',
 'data',
 'xenium_data',
 'Xenium_V1_Human_Ovary_tiny_outs.zip',
 'sample_data']
```

```
# Extract ZIP
```

```
import zipfile
```

```
zip_path = "/content/Xenium_V1_Human_Ovary_tiny_outs.zip"
```

```
with zipfile.ZipFile(zip_path, 'r') as zip_ref:
    zip_ref.extractall("/content/xenium_data")
```

```
print("Extraction completed")
```

Extraction completed

```
# Check Extracted Files
os.listdir("/content/xenium_data")
```

```
['gene_panel.json',
 'cells.parquet',
 'morphology.ome.tif',
 'analysis_summary.html',
 'transcripts.parquet',
 'nucleus_boundaries.csv.gz',
 'transcripts.zarr.zip',
 'morphology_focus',
 '.end-of-run',
 'cell_feature_matrix.tar.gz',
 'cells.zarr.zip',
 'cell_boundaries.parquet',
 'cell_feature_matrix.h5',
 'experiment.xenium',
 'analysis.tar.gz',
 'cell_feature_matrix.zarr.zip',
 'cell_boundaries.csv.gz',
 'cells.csv.gz',
 'metrics_summary.csv',
 'nucleus_boundaries.parquet',
 'aux_outputs.tar.gz',
 'analysis.zarr.zip']
```

```
import os
```

```
# Load Xenium expression matrix
```

```
import scanpy as sc

adata = sc.read_10x_h5("/content/xenium_data/cell_feature_matrix.h5")

adata
```

```
AnnData object with n_obs x n_vars = 632 x 403
var: 'gene_ids', 'feature_types', 'genome'
```

```
# Make gene names unique
```

```
adata.var_names_make_unique()
```

```
# Basic quality control
```

```
adata.obs["total_counts"] = adata.X.sum(axis=1)
adata.obs["n_genes_by_counts"] = (adata.X > 0).sum(axis=1)

adata.obs.head()
```

	total_counts	n_genes_by_counts
aaddnana-1	111.0	42
aagoiakf-1	139.0	44
abhgncbo-1	253.0	80
abijbcjf-1	122.0	53
ablcgmek-1	80.0	44

```
# Normalize and log-transform
```

```
sc.pp.normalize_total(adata)
sc.pp.log1p(adata)
```

```
# PCA, neighbors, UMAP
```

```
sc.pp.pca(adata)
sc.pp.neighbors(adata)
sc.tl.umap(adata)
```

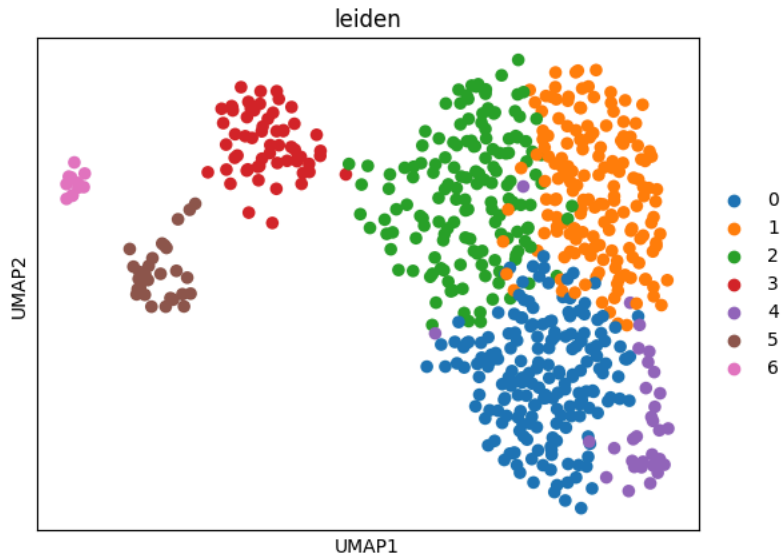
```
# Leiden clustering
```

```
sc.tl.leiden(adata)
```

```
/tmp/ipykernel_13391/3913797718.py:3: FutureWarning: The `igraph` implementation of leiden clustering is *orders of magnitude fa
In the future, the default backend for leiden will be igraph instead of leidenalg. To achieve the future defaults please pass: `
sc.tl.leiden(adata)
```

```
# UMAP visualization
```

```
sc.pl.umap(adata, color="leiden")
```



```
# Show available genes
```

```
print(adata.var_names[:30])
```

```
Index(['A2M', 'ACE2', 'ACTA2', 'ACTB', 'ADAM28', 'AIF1', 'AIRE', 'AKT1',
      'ANPEP', 'ANXA1', 'APOE', 'AREG', 'ARG1', 'ARID1A', 'ARPC3', 'ARPC5',
      'ATM', 'ATP1A1', 'BANK1', 'BATF3', 'BCL2', 'BRAF', 'C1QB', 'C1QBP',
      'C1R', 'C1S', 'CCL11', 'CCL13', 'CCL14', 'CCL15'],
      dtype='object')
```

```
# Example gene expression on UMAP
```

```
# Replace gene name if needed
```

```
sc.pl.umap(adata, color=adata.var_names[0])
```

